

The Lemke–Howson Algorithm

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We give a quick overview of the Lemke–Howson algorithm¹ algebraically. A simple 3×2 bimatrix game is provided to illustrate the complementary pivoting process and its computation.

¹ *Algorithmic Game Theory*. Cambridge University Press, 2007

Linear Complementary Problem

We consider a two-player $m \times n$ bimatrix game, where A is the payoff matrix for player 1 and B is the payoff matrix for player 2:

m : num of actions for player P_1 , n : num of actions for player P_2 .

We can then simplify our notations by defining a larger matrix C covering both payoff matrices:

$$C = \begin{pmatrix} 0 & A \\ B^T & 0 \end{pmatrix}, \quad (1)$$

which is required to be non-negative (and has no zero column). Its dimension is $d = m + n$. We then define:

$$S = \{z \in \mathbb{R}^d \mid z \geq 0, Cz \leq \mathbf{1}\}, \quad (2)$$

where $\mathbf{1}$ is a column vector with elements equal to 1. S is a polytope² by definition. The vector z can be considered as a (unnormalised) joint mixed policy:

$$z = \{x, y\} \in S = P \times Q;$$

$$z_i = x_i \text{ for } 1 \leq i \leq m, z_{m+j} = y_j \text{ for } 1 \leq j \leq n,$$

where P is the space bounded for player 1 and Q is for player 2. To make them as probabilities, we normalise them as:

$$\bar{x} = \frac{x}{\mathbf{1}^T x}, \bar{y} = \frac{y}{\mathbf{1}^T y}. \quad (3)$$

Search for an NE is equivalent to solving the following Linear Complementary Problem:

$$Cz + w = \mathbf{1}, \quad z \geq 0, \quad w \geq 0, \quad z_i w_i = 0 \text{ for } 1 \leq i \leq d, \quad (4)$$

where $w_i, 1 \leq i \leq d$ are slack variables. Typically we use *Complementary Pivoting* to algebraically solve it.

² A bounded intersection of a finite set of half-spaces.

We label $z = \{x, y\} \in S - \{0\}$ according to whether w_i is zero or z_i is zero³; and if $\{x, y\}$ is completely labeled from 1 to d and no duplication, we obtain a Nash equilibrium as $\{\bar{x}, \bar{y}\}$.

To understand how the Lemke-Howson algorithm works, we rewrite the constraints as follows:

$$w = 1 - Cz$$

where the left hand side of the equations contains *basic variables* and the right hand side contains *nonbasic variables*. The basic solution is obtained by setting all nonbasic variables to zero, so that the basic variables are equal to the constants. So the Lemke-Howson algorithm starts with the trivial solution $z = 0, w = 1$, which defines the *artificial equilibrium*: $\{x, y\} = 0$.

The algorithm works by a process called *pivoting*, that exchanges a nonbasic variable with a basic variable in such a way that the new dictionary defines the same linear relationship between its variables, and so that the corresponding basic solution remains feasible.

To visualise it, think of the solution as a corner (vertex) of a shape (polytope). Pivoting is like walking along the edge of this shape to move from one corner to the next as illustrated in Fig.1.

³ We assign a label i if w_i or z_i is zero. If both are zero, we have two i 's assigned - a duplicated case.

An Illustrative Example

Our running example is the following (as in Fig.1):

$$A = \begin{pmatrix} 0 & 6 \\ 2 & 5 \\ 3 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 4 & 3 \end{pmatrix}$$

$$m = 3, \quad n = 2, \quad d = m + n = 5$$

More specially, we have:

$$C = \begin{pmatrix} 0 & A \\ B^T & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & 3 & 3 \\ 1 & 0 & 4 & 0 & 0 \\ 0 & 2 & 3 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 6 & 2 \\ 4 & 3 \end{pmatrix}, \quad B^T = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 2 & 3 \end{pmatrix}$$

(basic variable ≥ 0)

Start with $z = 0, \quad w = 1$ as a solution to

$$w = 1 - Cz,$$

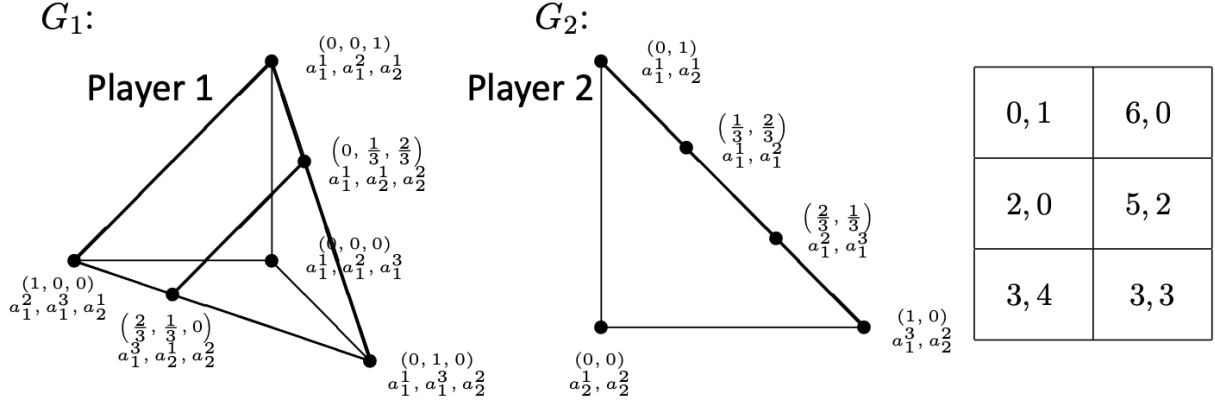


Figure 1: Probability simplex for Players 1 and 2's policies. player 2's strategy space is a triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$, while player 1's strategy space is a pyramid with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. It is convenient to add one fictitious point in the strategy space of each player, the origin; that is, $(0,0,0)$ for player 1 and $(0,0)$ for player 2. Each node is annotated with the mixed strategy to which it corresponds as well as the actions with which it is labeled.

gives

$$\begin{array}{rcl}
 w_1 = 1 & & -6z_5 \\
 w_2 = 1 & & -2z_4 \quad -5z_5 \\
 w_3 = 1 & & -3z_4 \quad -3z_5 \\
 w_4 = 1 & -z_1 & -4z_3 \\
 w_5 = 1 & -2z_2 & -3z_3
 \end{array}$$

Let us start with z_2 to enter the basis (left hand side) and w_5 to leave the basis. Rearranging the last equation gives

$$z_2 = \frac{1}{2} - \frac{1}{2}w_5 - \frac{3}{2}z_3.$$

So z_2 entering and w_5 leaving gives:

$$\begin{array}{rcl}
 w_1 = 1 & & -6z_5 \\
 w_2 = 1 & & -2z_4 \quad -5z_5 \\
 w_3 = 1 & & -3z_4 \quad -3z_5 \\
 w_4 = 1 & -z_1 & -4z_3 \\
 z_2 = \frac{1}{2} & -\frac{1}{2}w_5 & -\frac{3}{2}z_3
 \end{array}$$

In this case, the basic solution⁴ gives $z = \{0, 1/2, 0, 0, 0\} \Rightarrow \bar{x} = \{0, 1, 0\}, \bar{y} = \{0, 0\}$ after normalisation following Eq. 3. Geometrically, we have moved from the origin $z = \{x, y\} = \{0, 0, 0, 0, 0\} \Rightarrow \{\bar{x}, \bar{y}\} = \{0, 1, 0, 0, 0\}$.

As w_5 leaves the basic variables and enter the right hand side of the equation, becoming nonbasic variables (therefore zero). This means y_2 (index 5 of w is referred to the second action of player 2 so y_2) is the pure best response to x . As such, according to the condition of a Nash equilibrium, z_5 has to be assigned with positive probability

⁴ The basic solution is obtained by setting all nonbasic variables to zero, so that the basic variables are equal to the constants.

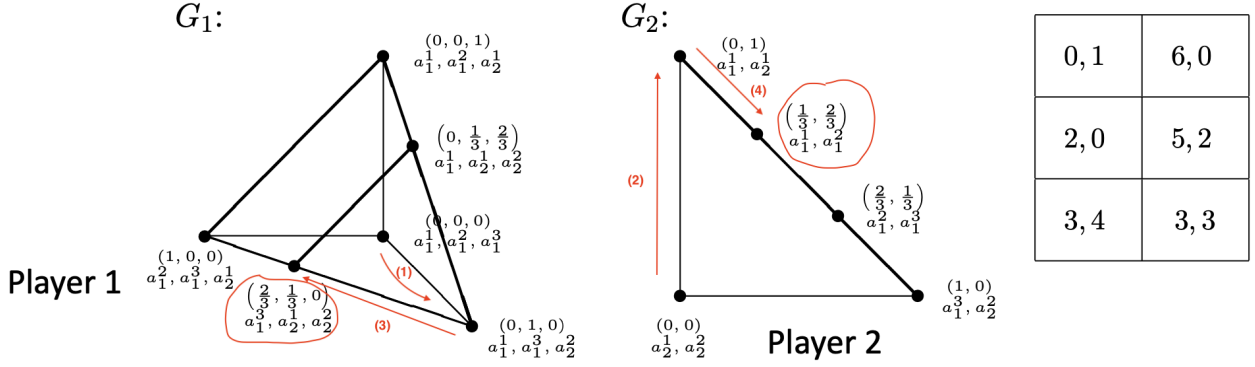


Figure 2: Complementary Pivoting, walking along the edges. The path is

$$\begin{aligned}
 &origin = \{0, 0, 0, 0, 0\} \\
 &\Rightarrow \{\bar{x}, \bar{y}\} = \{0, 1, 0, 0, 0\} \\
 &\Rightarrow \{\bar{x}, \bar{y}\} = \{0, 1, 0, 0, 1\} \\
 &\Rightarrow \{\bar{x}, \bar{y}\} = \{2/3, 1/3, 0, 0, 1\} \\
 &\Rightarrow \{\bar{x}, \bar{y}\} = \{2/3, 1/3, 0, 1/3, 2/3\}
 \end{aligned}$$

⁵ As z_5 deals with the second action of player 2, we are changing the policy of player 2.

(but it equals zero now). That is to say, we now, however, have z_5 and w_5 both as nonbasic variables - both are zero and are duplicated. We need one of them becomes zero but not all of them. This is due to the condition of a Nash equilibrium. So z_5 has to move to the left hand side of the equation, becoming a basic variable (this means assigning a positive probability)⁵.

So z_5 is allowed to increase while all other nonbasic variables staying at zero. The affected variables on the left hand side are

$$\begin{aligned}
 w_1 &= 1 & -6z_5 \\
 w_2 &= 1 & -5z_5 \\
 w_3 &= 1 & -3z_5,
 \end{aligned}$$

where we don't need to worry about z_4 just yet as it has been assigned to zero. When we increase the value of z_5 , we need to change only one of its w 's in the equation while also making sure the rest w remain non-negative (the slack variables have to be non-negative by construction):

$$\begin{aligned}
 w_1 &= 1 & -6z_5 \geq 0 \\
 w_2 &= 1 & -5z_5 \geq 0 \\
 w_3 &= 1 & -3z_5 \geq 0,
 \end{aligned}$$

where the stronger of these constraints is $w_1 = 1 - 6z_5 \geq 0 \Rightarrow z_5 = \frac{1}{6}$. Choosing the stronger constraint allow us to move w_1 to the right hand side, making $w_1 = 0$, while still keeping w_2 and w_3 positive. So we rearrange the first equation by $z_5 = \frac{1}{6} - \frac{1}{6}w_1$. We also replace $z_5 = \frac{1}{6} - \frac{1}{6}w_1$ for the second and the third equations:

$$\begin{array}{rclcl}
z_5 & = & \frac{1}{6} & & -\frac{1}{6}w_1 \\
w_2 & = & \frac{1}{6} & -2z_4 & \frac{5}{6}w_1 \\
w_3 & = & \frac{1}{2} & -3z_4 & \frac{1}{2}w_1 \\
w_4 & = & 1 & -z_1 & -4z_3 \\
z_2 & = & \frac{1}{2} & -\frac{1}{2}w_5 & -\frac{3}{2}z_3
\end{array}$$

So in this case, we have the sequence $z = \{x, y\} = \{0, 0, 0, 0, 0\} \Rightarrow z = \{x, y\} = \{0, 1/2, 0, 0, 0\} \Rightarrow z = \{x, y\} = \{0, 1/2, 0, 0, 1/6\} \Rightarrow \{\bar{x}, \bar{y}\} = \{0, 1, 0, 0, 1\}$. The last holds due to the normalisation. As we have both z_1 and w_1 on the right hand side, it is time for us to move z_1 into the basic variables (the left hand side):

$$\begin{array}{rclcl}
z_5 & = & \frac{1}{6} & & -\frac{1}{6}w_1 \\
w_2 & = & \frac{1}{6} & -2z_4 & \frac{5}{6}w_1 \\
w_3 & = & \frac{1}{2} & -3z_4 & \frac{1}{2}w_1 \\
z_1 & = & 1 & -w_4 & -4z_3 \\
z_2 & = & \frac{1}{2} & -\frac{1}{2}w_5 & -\frac{3}{2}z_3
\end{array}$$

This result in $z = \{x, y\} = \{0, 0, 0, 0, 0\} \Rightarrow z = \{x, y\} = \{0, 1/2, 0, 0, 0\} \Rightarrow z = \{x, y\} = \{0, 1/2, 0, 0, 1/6\} \Rightarrow z = \{x, y\} = \{1, 1/2, 0, 0, 1/6\} \Rightarrow \{\bar{x}, \bar{y}\} = \{2/3, 1/3, 0, 0, 1\}$. The last holds due to the normalisation from $\{x, y\} = \{1, 1/2, 0, 0, 1/6\}$. Since w_4 becomes a nonbasic variable, we need to turn z_4 into a basic variable (the left hand side). Again we have two equations related:

$$\begin{array}{rcl}
w_2 & = & \frac{1}{6} - 2z_4 \geq 0 \\
w_3 & = & \frac{1}{2} - 3z_4 \geq 0.
\end{array}$$

The first inequality gives the stronger constraint: $z_4 \leq \frac{1}{12}$. We thus replace $z_4 = \frac{1}{12} - \frac{1}{2}w_2 + \frac{5}{12}w_1$ with the second equation and use it to replace z_4 in the third equation, which gives

$$\begin{array}{rclcl}
z_5 & = & \frac{1}{6} & & -\frac{1}{6}w_1 \\
z_4 & = & \frac{1}{12} & -\frac{1}{2}w_2 & +\frac{5}{12}w_1 \\
w_3 & = & \frac{1}{4} & +\frac{3}{2}w_2 & -\frac{3}{4}w_1 \\
z_1 & = & 1 & -w_4 & -4z_3 \\
z_2 & = & \frac{1}{2} & -\frac{1}{2}w_5 & -\frac{3}{2}z_3.
\end{array}$$

Finally we reach to a solution. As we can see from the basic variables on the left hand side of the equations that the joint mixed strategy (unnormalised) is:

$$z = \{x, y\} = \{1, 1/2, 0, 1/12, 1/6\}$$

Normalising it gives:

$$\{x, y\} = \{2/3, 2/3, 0, 1/3, 2/3\}$$

It is completely labelled as $z_i w_i = 0$ for $1 \leq i \leq d$ holds.

The path is

$$\begin{aligned} origin &= \{0, 0, 0, 0, 0\} \\ \Rightarrow \{\bar{x}, \bar{y}\} &= \{0, 1, 0, 0, 0\} \\ \Rightarrow \{\bar{x}, \bar{y}\} &= \{0, 1, 0, 0, 1\} \\ \Rightarrow \{\bar{x}, \bar{y}\} &= \{2/3, 1/3, 0, 0, 1\} \\ \Rightarrow \{\bar{x}, \bar{y}\} &= \{2/3, 1/3, 0, 1/3, 2/3\} \end{aligned}$$

References

Algorithmic Game Theory. Cambridge University Press, 2007.